

Ritesh Khatkar

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PROFESSIONAL SUMMARY

Computer Science student specializing in Artificial Intelligence and Machine Learning (AI/ML) with hands-on experience in computer vision, deep learning, and intelligent automation. Proven track record of developing AI-driven security systems and enterprise solutions that reduced manual workload by 70% and improved operational efficiency by 35%. Proficient in Python, TensorFlow, and OpenCV with expertise in deploying real-time ML models for practical applications.

TECHNICAL SKILLS

- Programming Languages:** Java, Python, HTML, CSS, SQL
- AI & Machine Learning:** Machine Learning (ML), Deep Learning, Computer Vision, Natural Language Processing (NLP), Supervised & Unsupervised Learning, Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Transformers, YOLO (You Only Look Once), Retrieval-Augmented Generation (RAG)
- Tools & Frameworks:** TensorFlow, Keras, Scikit-learn, OpenCV, LangChain, Streamlit, Pandas, NumPy, Matplotlib, Flask
- Data Science & Analytics:** Tableau, Power BI, Data Visualization, Statistical Analysis
- Technologies:** Git, GitHub, Blockchain, Zero-Knowledge Proof (ZKP), RFID, QR Codes, REST APIs
- Specialized Areas:** Financial Analysis, Computer Vision, Biometric Security, Database Management

PROFESSIONAL EXPERIENCE

- Airport Authority of India, Kolhapur** July 2024 – August 2024
AI/ML Intern Kolhapur, India
 - Designed and deployed an AI-driven Unattended Baggage Detection System using OpenCV and machine learning to monitor 15+ CCTV feeds in real-time, reducing manual surveillance workload by 70%.
 - Trained computer vision models on 500+ hours of surveillance footage achieving 90.6% detection accuracy and reducing false alarms by 40%.
 - Proposed system integration recommendations that could reduce security response times by 30%, recognized by senior airport officials.
- Softech Solutions Pvt. Ltd.** June 2024 – August 2024
Software Development Intern Kolhapur, India
 - Developed centralized database management system for sugar manufacturers, automating inventory tracking, procurement, and order processing, reducing manual labor by 35% and increasing supply chain efficiency by 25%.
 - Implemented RFID tagging and QR code-based farmer order management system improving traceability for 5,000+ sugarcane orders and reducing data entry errors by 60%.
 - Analyzed 6 months of operational data to identify bottlenecks causing 15% production delays and recommended system optimizations for digital transformation of agri-supply chains.

EDUCATION

- Kolhapur Institute of Technology** 2022 – 2026
Bachelor of Engineering in Computer Science (AI & ML Specialization) Kolhapur, India
 - CGPA: 7.27/10.00
- Shri Anandrao Abitkar Junior College** 2020 – 2022
Pre-University Education, Grade: 71% Gargoti, India
 - Percentage: 71.17%
- Shri Shahu Kumar Bhavan** 2019 – 2020
Secondary Education, Grade: 81% Gargoti, India
 - Percentage: 81.0%

HACKATHON PROJECTS

• Exhibit Monitor – Text File Processing Application

2025

Zensar Technologies Hackathon

- Developed robust backend application for automated text file monitoring and processing, handling 100+ daily files up to 10 GB with real-time folder monitoring every 30 seconds.
- Engineered comprehensive validation system checking file name validity, arrival time compliance, and duplicate detection with automatic deletion for failed validations.
- Implemented JSON-based configuration system enabling format changes without code modifications and error logging for debugging and audit purposes.

• E-Governance Management System

2025

Quantbit Technologies Hackathon

- Built comprehensive web-based e-governance platform connecting citizens with municipal officers for civic issue reporting and resolution.
- Implemented role-based access control system with real-time status tracking dashboard displaying issue progress and resolution timelines.
- Designed scalable multi-city architecture with automated issue categorization, priority assignment, and notification system.

KEY PROJECTS

• Virtual Reality Campus Navigation System

Unreal Engine, Blender, Blueprints, Git

- Developed immersive VR campus exploration simulation used by 150+ prospective students featuring 360° virtual tours with LoD4 3D realism and real-world architectural mapping.
- Modeled 20+ campus structures using high-fidelity LoD4 standards with three navigation modes (eagle view, third-person, first-person), increasing spatial awareness by 60%.
- Integrated robot-guided tours and gamified interactions including quest objects and trivia checkpoints, resulting in 45% increase in user engagement during beta testing.

• Zero-Knowledge Proof Verified IRIS Biometric System

Python, TensorFlow, Scikit-learn, OpenCV, Flask, Blockchain, ZKP

- Developed biometric authentication system using Zero-Knowledge Proof (ZKP) to verify IRIS dataset with 50,000+ samples ensuring 100% privacy and zero data leakage.
- Implemented three innovative ZKP protocols with blockchain-inspired verification mechanisms, reducing verification time by 35% and achieving 99.9% data integrity accuracy.
- Achieved 97.8% match precision for iris recognition algorithms with cryptographic validation across five simulated nodes, proving scalable, decentralized, and privacy-compliant architecture.

CERTIFICATIONS

• AI For Everyone – DeepLearning.ai

September 2023

• Supervised Machine Learning: Regression and Classification – DeepLearning.ai

January 2024

• Generative AI with Large Language Models – DeepLearning.ai

August 2024

• Deloitte Australia – Data Analytics Job Simulation

June 2024

• Deloitte Australia – Technology Job Simulation

June 2024

LEADERSHIP & VOLUNTEERING

• Event Volunteer

February 2023

E-Summit'23, KIT College of Engineering

- Managed attendee registration and maintained event discipline for college entrepreneurship summit.

ADDITIONAL INFORMATION

Languages: English (Fluent), Hindi (Fluent), Marathi (Native), Kannada (Conversational)

Interests: Gaming, Sports, Financial Analysis, Data Science, Travel, Reading Financial Articles